



JOMO KENYATTA UNIVERSITY OF AGRICULTURE & TECHNOLOGY

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elearning@jkuat.ac.ke

BIT 2223 Mobile Computing

LAST REVISION ON September 25, 2013





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



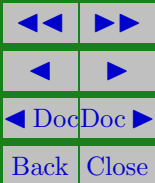
LESSON 4

Wireless Networks

Objectives

By the end of this topic you should be able to;

- Explain routing in ad-hoc wireless networks
- Explain various routing protocols as used in ad-hoc network.
- Give various market place growth and dominant technologies





4.1. Introduction

Wireless data networks exist in such number and variety as to be difficult to categorize and compare.

Some wireless data networks run over wireless voice networks, such as mobile telephone networks. CPDP, HSCSD, PDC-P, and GPRS are examples. Other wireless networks run on their own physical layer networks, utilizing anything from antennas built into hand held devices to large antennas mounted on towers. 802.11, LMDS, and MMDS are examples. A few wireless networks are intended only to connect small devices over short distances. Blue-tooth is an example.

Wireless networks which run over other wireless networks often utilize the lower layer networks to provide security and encryption. Stand-alone wireless networks either provide their



BIT 2223 Mobile Computing

own security and encryptions features or rely upon VPN's (Virtual Private Networks) to provide those features. In many cases, multiple layers of security and encryption may be desirable.

Some wireless networks are fixed, meaning that antennas do not move frequently. Other wireless networks are mobile, meaning that the antenna can move constantly. This is sometimes a feature of the specific implementation and antenna design, instead of an inherent limitation of the wireless network specification.

Wireless networks may operate on licensed or unlicensed portions of the frequency spectrum.

4.2. Common Wireless Network Types

Major types of wireless networks include:



CDPD	Cellular Digital Packet Data
HSCSD	High Speed Circuit Switched Data
PDC-P	Packet Data Cellular
GPRS	General Packet Radio Service
1xRTT	1x Radio Transmission Technology
Blue-tooth	
IrDA	Infrared
MMDS	Multichannel Multi-point Distribution Service
LMDS	Local Multi-point Distribution Service
WiMAX	Worldwide Interoperability for Microwave Access
802.11	Wi-Fi



4.3. Routing in ad-hoc wireless networks

A **mobile ad-hoc network** (MANET), sometimes called a mobile mesh network, is a self-configuring network of mobile devices connected by wireless links. MANETs are a kind of wireless ad-hoc networks that usually has a routable networking environment on top of a Link Layer ad-hoc network.

Each device in a MANET is free to move independently in any direction, and will therefore change its links to other devices frequently. Each must forward traffic unrelated to its own use, and therefore be a router. The primary challenge in building a MANET is equipping each device to continuously maintain the information required to properly route traffic.

An **ad-hoc routing protocol** is a convention, or standard, that controls how nodes decide which way to route packets be-



tween computing devices in a mobile ad-hoc network.

In ad-hoc networks, nodes do not start out familiar with the topology of their networks; instead, they have to discover it. The basic idea is that a new node may announce its presence and should listen for announcements broadcast by its neighbors. Each node learns about nodes nearby and how to reach them, and may announce that it, too, can reach them

The following is a list of some ad-hoc network routing protocols:

4.3.1. Pro-active (table-driven) routing

This type of protocols maintains fresh lists of destinations and their routes by periodically distributing routing tables throughout the network. The main disadvantages of such algorithms



are:

- Respective amount of data for maintenance.
- Slow reaction on restructuring and failures.

4.3.2. Reactive (on-demand) routing

This type of protocols finds a route on demand by flooding the network with Route Request packets. The main disadvantages of such algorithms are -

- High latency time in route finding.
- Excessive flooding can lead to network clogging.

4.3.3. Flow-oriented routing

This type of protocols finds a route on demand by following present flows. One option is to unicast consecutively when for-



warding data while promoting a new link. The main disadvantages of such algorithms are

- Takes long time when exploring new routes without prior knowledge.
- May refer to entitative existing traffic to compensate for missing knowledge on routes.

4.3.4. Adaptive (situation-aware) routing

This type of protocols combines the advantages of proactive and of reactive routing. The routing is initially established with some proactively prospected routes and then serves the demand from additionally activated nodes through reactive flooding. Some metrics must support the choice of reaction. The main disadvantages of such algorithms are -



- Advantage depends on amount of nodes activated.
- Reaction to traffic demand depends on gradient of traffic volume.

4.3.5. Hybrid (both pro-active and reactive) routing

This type of protocols combines the advantages of proactive and of reactive routing. The routing is initially established with some proactively prospected routes and then serves the demand from additionally activated nodes through reactive flooding. The choice for one or the other method requires predetermination for typical cases. The main disadvantages of such algorithms are -

- Advantage depends on amount of nodes activated.
- Reaction to traffic demand depends on gradient of traffic



volume.

4.3.6. Hierarchical routing protocols

With this type of protocols the choice of proactive and of reactive routing depends on the hierarchic level where a node resides. The routing is initially established with some proactively prospected routes and then serves the demand from additionally activated nodes through reactive flooding on the lower levels. The choice for one or the other method requires proper attribution for respective levels. The main disadvantages of such algorithms are

- Advantage depends on depth of nesting and addressing scheme.
- Reaction to traffic demand depends on meshing param-



ters.

4.3.7. Geographical routing protocols

This type of protocols acknowledges the influence of physical distances and distribution of nodes to areas as significant to network performance. The main disadvantages of such algorithms are

- Efficiency depends on balancing the geographic distribution versus occurrence of traffic.
- Any dependence of performance with traffic load thwarting the negligence of distance may occur in overload.

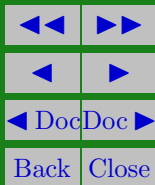


4.3.8. Power-aware routing protocols

Energy required to transmit a signal is approximately proportional to d^α , where d is the distance and α is the attenuation factor or path loss exponent, which depends on the transmission medium. When $\alpha = 2$ (which is the optimal case), transmitting a signal half the distance requires one fourth of the energy and if there is a node in the middle willing spend another fourth of its energy for the second half, data would be transmitted for half of the energy than through a direct transmission - a fact that follows directly from the inverse square law of physics.

The main disadvantages of such algorithms are

1. This method induces a delay for each transmission.
2. No relevance for energy network powered transmission op-





erated via sufficient repeater infrastructure.

Intelligent agent

Intelligent agent technology provides a mechanism for information systems to act on behalf of their users. They can be programmed to search, acquire, and store information on behalf of the wants and needs of users.

4.4. Attributes

- Intelligent agents are task-oriented. Examples of intelligent agent tasks include data mining, profile management, privacy management, rules management, and application management.
- Data mining agents seek data and information based on



BIT 2223 Mobile Computing

the profile of the user and instructions carried out by the rules manager agent.

- The profile management agent's role is to maintain the user profile, constantly adding, deleting and modifying profile information based on new information.
- The privacy management agent's role is to safeguard privacy of the user, including identity, location, or other personal information.
- An application agent manages application(s) and interactions between the network, other networks, content, and other applications.



4.4.1. Agent Locations and Technologies

Intelligent agent software may consist of embedded technology within the mobile device, servers on the Internet, and/or programs within applications on/off the mobile network.

4.4.2. Agency vs. Intelligence

Agency is the degree of authority given to a program - how much it is allowed to perform on its own accord, based on the instructions given by user.

Intelligence is the degree of reasoning and independent "learning" enabled within the program. A truly intelligent agent has the ability to adapt itself to its environment, sensing events and objects, evaluating circumstances and objects, and act over time in pursuit of a specific agenda.



4.4.3. Intelligent Agent Applications

The numbers of applications benefiting from intelligent agent technology are virtually limited only by the imagination. One of the key goals is to enable the autonomous delivery of information for the benefit of various applications. For example, a location detection intelligent agent in a mobile device would inform application middle-ware once a user is in a certain location, perhaps engaging additional positioning and or applications.

4.4.4. Market Place Growth And Dominant Technologies.

The Information Technology (IT) Revolution is driven by four interlinked core industries: computers, semiconductors, software and telecommunications. These industries are built around the



technologies of:

- Microelectronics (the design and fabrication of integrated circuits);
- Software (the instruction sets that tell the circuits what to do); and
- Wireless and Photonics (the generation, control and detection of EM radiation, for example, to transmit and receive signals).

This interconnected cluster generates widespread innovation. It creates new products like Blackberries and GPS location. The trigger for these major eras of technology-driven change is a breakthrough innovation that opens up a whole new frontier of design and product possibilities, sparking the imagination of engineers, entrepreneurs and investors.



BIT 2223 Mobile Computing

EXERCISE 1.  Explain MANET.

Example . Compare and contrast agency and intelligence.

Solution: Agency is the degree of authority given to a program - how much it is allowed to perform on its own accord, based on the instructions given by user. Intelligence is the degree of reasoning and independent "learning" enabled within the program.



EXERCISE 2.  List and explain any two network routing protocols.



Solutions to Exercises

Exercise 1. A mobile ad-hoc network (MANET) is a self-configuring network of mobile devices connected by wireless links.

Exercise 1



JKUAT SODeL
©2013

BIT 2223 Mobile Computing

Exercise 2. i) Flow-oriented routing; ii) Hierarchical routing protocols

Exercise 2

